

Intelligent Elephant Care and Visitor Insights

Project ID: 25-26J-391

Project Proposal Report

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B.Sc. (Hons) in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

June 2025

**Visitor Data Analytics for Optimized Experience
and Elephant Welfare Analyzing visitor**

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The Pinnawala Elephant Orphanage is one of Sri Lanka's most significant tourist destinations, attracting over 40,000 visitors per month. While this high visitation level contributes to national revenue and global awareness of elephant conservation, it also creates substantial challenges in crowd management, visitor satisfaction, and animal welfare. Overcrowding, long waiting times, and poorly managed visitor flow often reduce the quality of the tourist experience while simultaneously exerting stress on the elephants. The proposed research addresses these challenges by designing a visitor data analytics framework that monitors, evaluates, and predicts visitor behavior patterns in real-time. The system will integrate IoT-based visitor tracking, survey feedback, and historical datasets to identify congestion zones, dwell times, and behavioral trends. These insights will then be correlated with elephant welfare indicators, such as stress signals and movement disruptions, to identify visitor-induced stress triggers. Predictive models and real-time dashboards will be developed to optimize visitor flows, improve facility planning, and ensure proactive welfare measures. The proposed system aligns with the goals of sustainable and ethical tourism, offering practical tools for sanctuary managers to balance revenue generation, conservation, and visitor satisfaction. By introducing a cross-domain integration of visitor analytics and animal welfare, this study provides a novel contribution that can be applied to other conservation sites globally.

Keywords- Visitor Data Analytics , Experience Optimization , Elephant Welfare , IoT-based Tracking , Predictive Modeling , Real-time Dashboards , Ethical Tourism , Crowd Management , Conservation Technology , Pinnawala Elephant Orphanage

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List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
IOT	Internet of Things
UX	User Experience
KPI	Key Performance Indicator
PEO	Pinnawala Elephant Orphanage

1. INTRODUCTION

1.1 Background

Tourism is a vital sector in Sri Lanka, contributing significantly to the economy and supporting biodiversity conservation initiatives. Among the country's most visited destinations is the Pinnawala Elephant Orphanage, a sanctuary established to protect orphaned and injured elephants. The site is internationally recognized for its dual purpose: ensuring the welfare of elephants while providing an educational and recreational space for visitors. However, the increasing number of tourists—reaching tens of thousands per month—has led to pressing management challenges. The orphanage must not only accommodate visitor expectations but also protect the elephants from stress induced by excessive human presence, noise, and overcrowding. Traditional management approaches such as manual monitoring, ticket-based scheduling, and basic crowd control have proven insufficient in addressing these issues. [1]

Recent advancements in data analytics, IoT, and artificial intelligence offer innovative ways to tackle these challenges. Visitor behavior can now be tracked through digital systems such as Wi-Fi hotspots, mobile applications, and smart cameras, while real-time analytics can provide actionable insights into footfall distribution and crowding hotspots. At the same time, conservation science emphasizes the importance of correlating human activity with animal welfare indicators. Thus, integrating visitor analytics with welfare monitoring has the potential to establish a holistic approach to sanctuary management, balancing conservation ethics with visitor experience. [2]

1.2 Literature Review

A wide body of literature has examined the role of technology in wildlife conservation and visitor management. Panicker & Joseph (2020) demonstrated the use of machine learning in tracking elephants through GPS data, showing how predictive analytics can support conservation. Khatri et al. (2022) reviewed AI-based behavior analysis in wildlife, highlighting its role in stress and activity monitoring. Studies in zoos and parks, such as Lund et al. (2024), used deep learning to analyze elephant behavior, while Kumar et al. (2019) assessed non-invasive stress indicators in captive elephants. On the tourism side, Dayananda & Weerakoon (2012) conducted surveys at PEO, revealing how overcrowding directly impacts visitor satisfaction and perceptions of animal welfare. [3]

Despite these contributions, most existing research treats animal welfare and visitor management as separate domains. Smart tourism studies focus on optimizing visitor flow and improving services but rarely integrate animal stress responses. Similarly, conservation research prioritizes animal monitoring but overlooks the influence of visitor behavior. A limited number of interdisciplinary works suggest that human presence can affect animal stress, yet there is no comprehensive framework to address both domains simultaneously at high-density tourist sites. This leaves a significant opportunity for a data-driven system that aligns visitor satisfaction with conservation ethics. [4]

1.3 Research Gap

Current practices at PEO do not effectively integrate visitor data analytics with elephant welfare indicators. Visitor management is reactive, relying on ticketing and basic crowd control, while welfare monitoring depends on manual observation. This lack of predictive and integrated tools leads to inefficiencies, overcrowding, and avoidable stress for elephants. Furthermore, there is no decision-support platform that provides staff with realtime insights into both visitor trends and animal welfare impacts. This research seeks to address this gap by developing a cross-domain analytics framework. By correlating visitor data (e.g., movement patterns, dwell times, feedback) with elephant behavior data, the proposed system will proactively identify and mitigate risks, thereby creating a sustainable and ethical tourism model. [5]

Research Gap	Existing Research	Identified Gap	Proposed Research Contribution
Visitor Management	Current practices at PEO rely on ticketing, scheduling, and manual crowd observation.	No predictive or realtime data-driven system to manage visitor flows.	Develop real-time analytics & predictive models for crowd forecasting.
Visitor Behavior Analytics	Studies have focused on surveys (e.g., Dayananda & Weerakoon, 2012) but mostly provide static insights.	Lack of continuous monitoring of visitor movement, dwell time, and engagement.	Implement IoT-based tracking and data analytics to monitor behavior dynamically.

Integration with Animal	Animal monitoring studies (Kumar et al., 2019; Lund et al., 2024) focus on stress detection but ignore visitor influence.	No framework correlating visitor activity with elephant welfare.	Cross-domain framework linking visitor data with welfare indicators.
Technology Use	AI/ML used in elephant tracking and stress analysis (Panicker & Joseph, 2020; Khatri et al., 2022).	Applied mainly in animal monitoring, not visitor flow or experience optimization.	Apply ML for forecasting high-traffic periods and identifying welfare risks.
Tourism Experience	Smart tourism studies optimize services but neglect ethical animal interactions.	Visitor satisfaction and animal welfare treated separately.	Combine experience optimization with conservation-friendly tourism practices.
Decision Support	No centralized tool for staff to make data-driven operational decisions.	Staff rely on manual judgment without realtime alerts or dashboards.	Develop dashboards & alert systems for proactive crowd and welfare management.

Table 1 : Research Gap

1.4 Research Problem

The Pinnawala Elephant Orphanage attracts more than 40,000 visitors per month, making it one of Sri Lanka's most visited wildlife sanctuaries. While this provides significant economic and conservation benefits, it also creates critical challenges in balancing visitor satisfaction with elephant welfare. Current visitor management practices rely on manual observation, ticketing, and basic scheduling systems, which do not provide adequate insights into visitor flow, crowd density, or engagement levels. As a result, overcrowding, long waiting times, and uneven distribution of visitors across exhibits frequently occur, reducing the quality of the tourist experience. [6]

More importantly, unmanaged visitor presence can negatively affect elephant welfare. Elephants exposed to excessive human interaction, noise, and crowding may display stress-related behaviors such as agitation, withdrawal, or changes in movement patterns. However, existing systems at PEO lack mechanisms to correlate visitor activity with these welfare indicators. Visitor management and elephant monitoring remain separate processes, leading to a reactive rather than proactive approach to welfare protection. [7]

Although studies in smart tourism and wildlife monitoring highlight the potential of IoT, data analytics, and machine learning for optimizing human–animal interactions, there is currently no integrated framework that applies these technologies to both visitor experience and elephant welfare in high-density tourist environments. The absence of such a system prevents sanctuary managers from making data-driven, real-time decisions that enhance visitor engagement while safeguarding elephants. [8]

Therefore, the research problem is the **lack of a comprehensive, data-driven visitor analytics framework that integrates visitor behavior with elephant welfare indicators, enabling proactive decision-making for crowd management and ethical tourism at Pinnawala Elephant Orphanage.**

2 OBJECTIVES

2.1 Main Objective

To design and implement a data-driven visitor analytics framework that optimizes visitor experience and supports proactive elephant welfare management at Pinnawala Elephant Orphanage.

2.2 Specific Objectives

- To collect and analyze visitor behavior data, including movement patterns, dwell times, and survey feedback, using IoT-enabled tracking and mobile applications.
- To integrate visitor data with elephant behavioral and stress indicators to identify visitor-induced stress triggers.
- To develop predictive models that forecast visitor peaks and potential welfare risks.
- To design real-time dashboards and alert systems for staff to manage visitor flow and optimize park operations.
- To promote sustainable and ethical tourism by aligning visitor satisfaction with animal welfare.

3 METHODOLOGY

3.1 Requirement Analysis

The project begins with a requirement analysis phase that identifies stakeholders (park staff, management, visitors, and conservation experts) and gathers their expectations. Visitor data requirements include footfall counts, path tracking, and dwell times, which will be collected through IoT sensors (e.g., Wi-Fi or Bluetooth trackers), CCTV cameras, and mobile applications. Visitor feedback will be collected using surveys and app-based forms. Elephant welfare data, such as movement disruptions and stress calls, will be integrated from parallel system components within the research group. Analytical requirements include real-time processing, visualization tools, and predictive modeling capabilities.

3.2 Feasibility Study

The feasibility study covers four dimensions:

- **Technical feasibility:** IoT sensors, cloud-based data analytics platforms, and visualization tools are widely available and can support the proposed framework.
- **Operational feasibility:** Park staff can easily use dashboards to monitor visitor flows and take preventive measures in real-time.
- **Economic feasibility:** The cost of implementing IoT sensors and data processing is moderate compared to the long-term benefits of improved visitor satisfaction and sustainable tourism.
- **Legal/Ethical feasibility:** Visitor privacy will be ensured through anonymization of data, while animal welfare guidelines will be respected to prevent intrusive monitoring.

4 Implementation

The implementation phase involves several steps. First, data collection mechanisms will be deployed across visitor zones, integrating IoT devices, mobile apps, and surveys. Data will be preprocessed to handle missing values and ensure consistency. Machine learning techniques will then be applied, including clustering (e.g., K-Means) to identify visitor groups and time-series forecasting (e.g., ARIMA, LSTM) to predict high-traffic periods. Correlation models will link visitor density with elephant stress data to identify welfare risks. Finally, dashboards and alert systems will be developed using visualization tools such as Power BI, Tableau, or custom-built web applications.

5 System Architecture

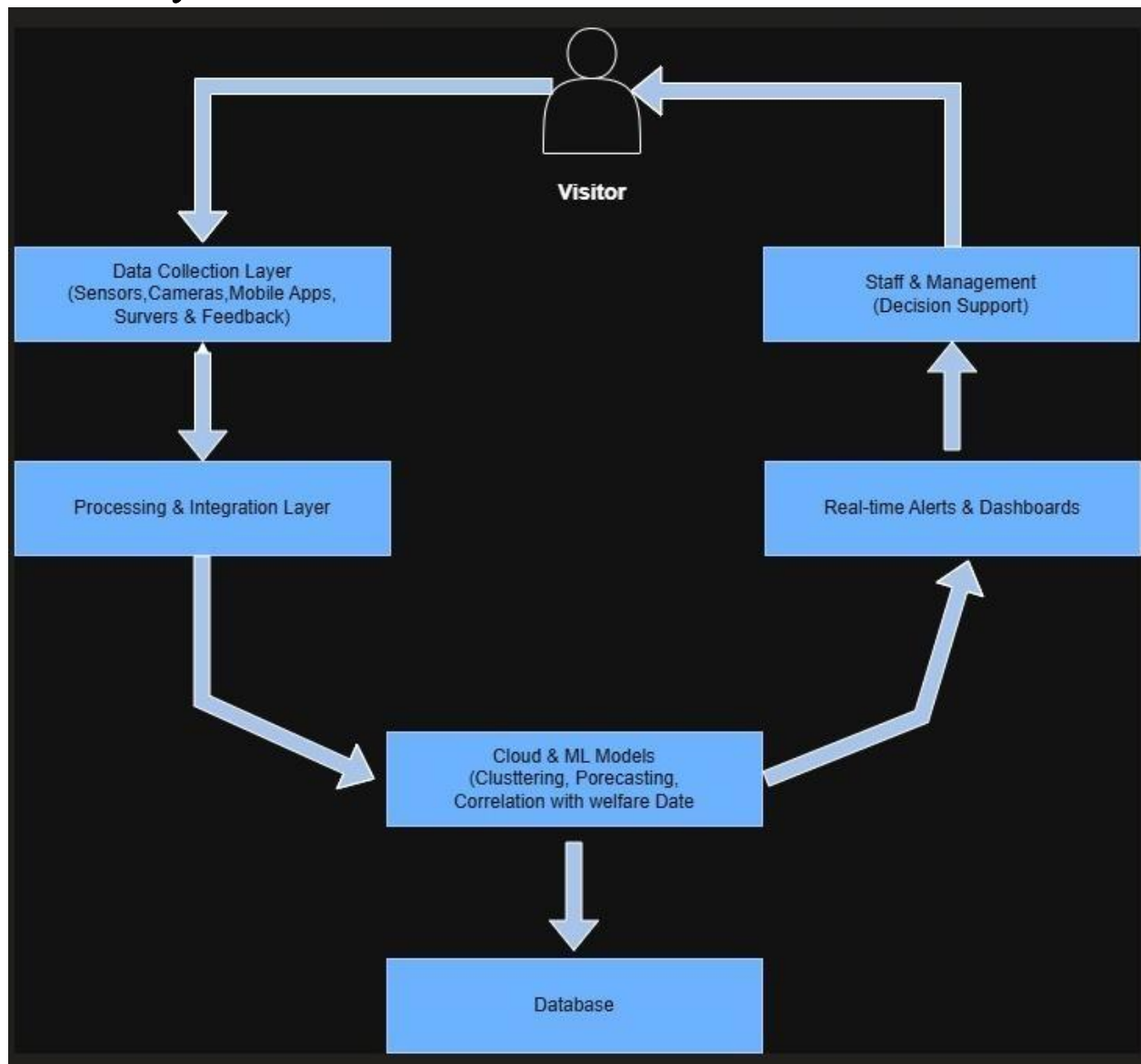
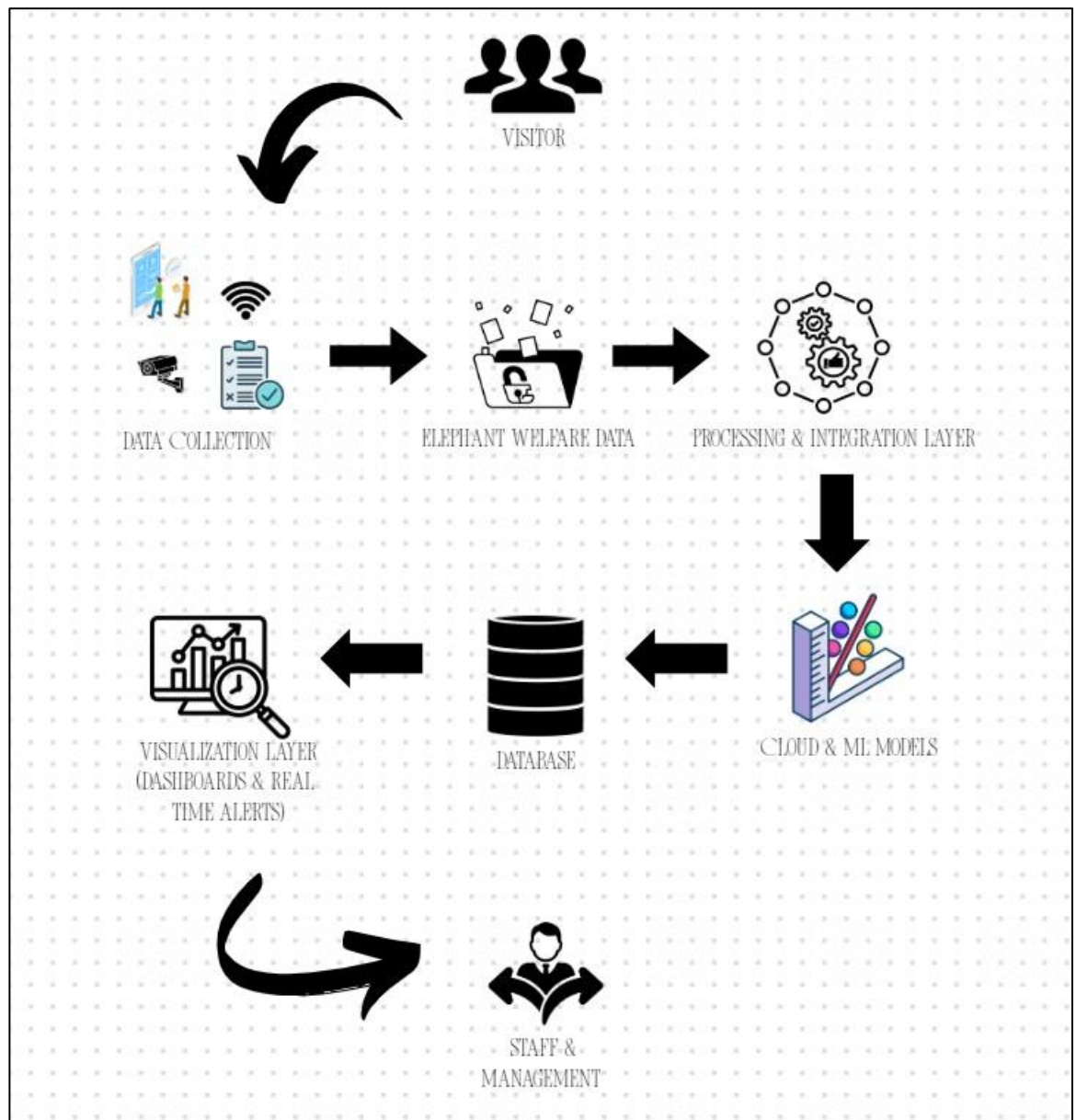


Figure 1 : System Architecture

6 System Diagram



7 PERSONNEL AND COMMERCIALIZATION

7.1 PERSONNEL

A focus on data analytics and visitor management systems. The researcher will be responsible for the design, development, and implementation of the visitor analytics framework, including requirement analysis, data collection, model development, and system testing. Guidance and supervision will be provided by the appointed academic supervisor, who will ensure that the research aligns with both technical and academic standards.

In addition, collaborative support will be sought from the staff of Pinnawala Elephant Orphanage to facilitate access to visitor data and provide domain-specific insights into operational challenges and welfare concerns. Technical inputs from other group members working on elephant health monitoring, diet recommendations, and stress detection will also be integrated into the system to ensure that visitor analytics is correlated with animal welfare indicators. The project may also benefit from consultations with external experts in tourism management and wildlife conservation to validate the ethical and practical implications of the proposed system. Collectively, these contributions will enable the successful implementation of a real-world, data-driven solution.

7.2 Commercialization

The proposed visitor analytics framework has strong commercialization potential both within Sri Lanka and internationally. At the local level, the system can be directly adopted by the Pinnawala Elephant Orphanage to enhance visitor experience, improve crowd

management, and strengthen its reputation as a sustainable and ethical tourism destination. This will contribute not only to higher visitor satisfaction and revenue but also to improved global branding of Sri Lanka's wildlife tourism sector.

Beyond Pinnawala, the system can be scaled and adapted to other wildlife parks, zoos, and nature reserves that face similar challenges in balancing visitor engagement with animal welfare. The framework can be developed as a **modular software-as-a-service (SaaS) platform**, enabling multiple institutions to customize it based on their needs. By incorporating analytics dashboards, predictive modeling, and mobile-based visitor engagement features, the system can attract interest from international conservation organizations and eco-tourism operators.

Furthermore, commercialization aligns with the growing demand for **smart tourism and conservation technologies**. As global tourists increasingly value ethical and sustainable practices, sanctuaries and parks adopting such systems will gain competitive advantage. Partnerships with government tourism authorities, NGOs, and private operators can expand deployment, while potential revenue streams may include subscription-based services, licensing, or consultancy packages. Ultimately, the commercialization of this research will support the dual goals of **conservation-friendly tourism and technological innovation**, ensuring both social and economic impact.

8 WORK BREAKDOWN STRUCTURE

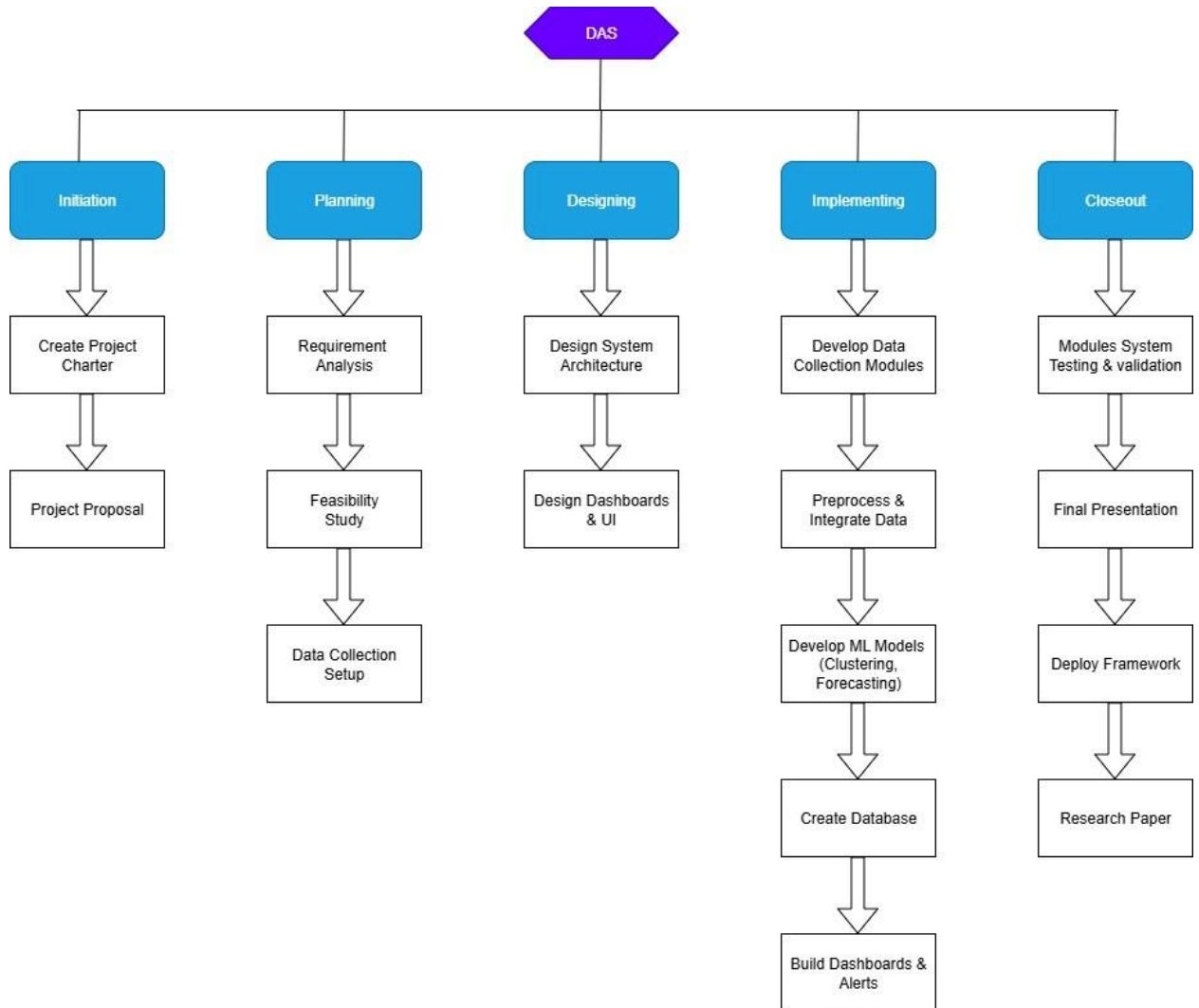


Figure 3 : Work Breakdown

9 GANTT CHART

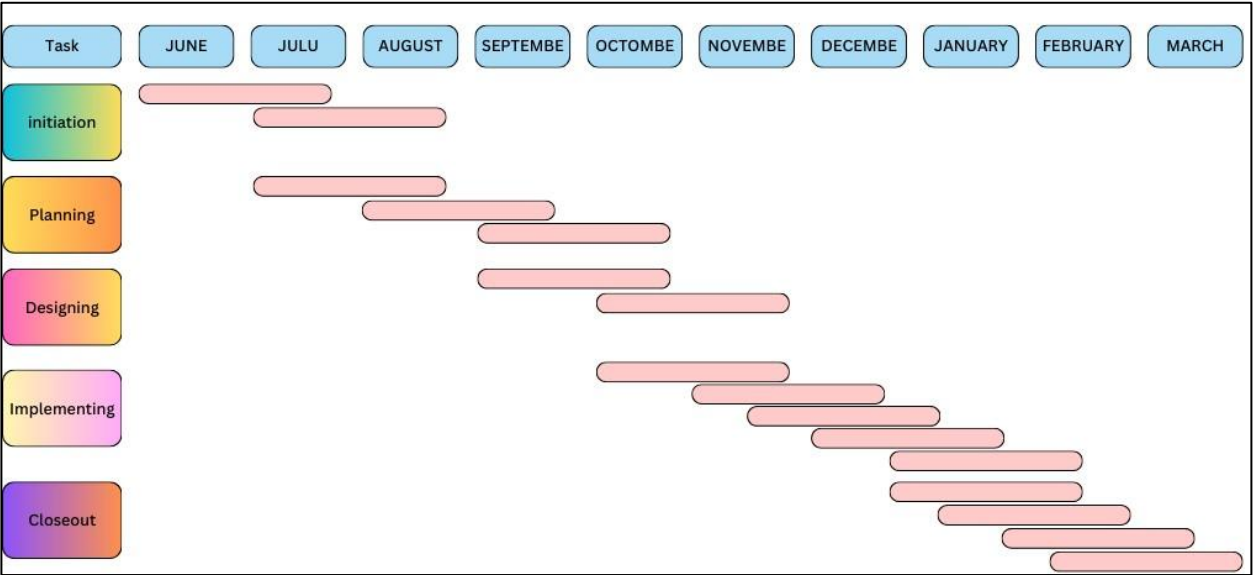


Figure 4 : Gant Chart

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References

- [1] D. & J. S. Panicker, AI in wildlife conservation: A case study on elephant tracking using machine learning., (2020).
- [2] P. S. R. & G. V. Khatri, "Use of AI for behavior analysis in wildlife: A review. Ecological Informatics, 68, 101510.," <https://doi.org/10.1016/j.ecoinf.2022.101510>, 2020.
- [3] S. M. O. C. & E. K. L. Lund, "Behavioral coding of captive African elephants using deep learning. Animals, 14(19), 2820.," <https://doi.org/10.3390/ani14192820>, 2024.
- [4] V. B. N. & R. T. Kumar, "Non-invasive assessment of physiological stress in captive Asian elephants. Animals, 9(8), 553.," <https://doi.org/10.3390/ani9080553>, (2019).
- [5] B. & W. D. Dayananda, "Visitor survey of the Pinnawala Elephant Orphanage, Sri Lanka. Gajah, 37, 3–10.," (2012).
- [6] M. & S. K. Dhairykar, " Management of nutrition in captive Asian elephants. International Journal of Veterinary Science & Animal Husbandry, 5(4), 45–51.," (2020).
- [7] P. & P. J. Fernando, "Range-wide status of Asian elephants. Gajah, 35, 15–20.," (2011).
- [8] B. A. Schulte, "Social structure and helping behavior in captive elephants. Zoo Biology, 19(5), 447–459.," [https://doi.org/10.1002/1098-2361\(2000\)19:5](https://doi.org/10.1002/1098-2361(2000)19:5), (2000).